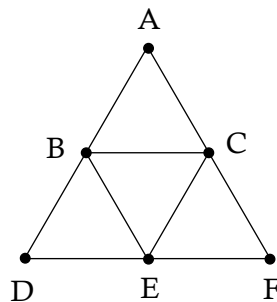
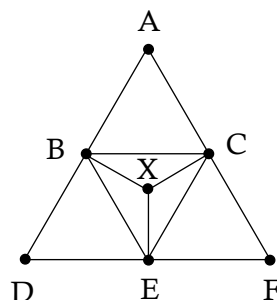


1. A simple graph has set of vertices $\{1, 2, 3, 4, 5\}$. Two vertices are adjacent in the graph if and only if their sum is a prime number.
 - a. Draw this graph.
 - b. What is the order of this graph? What is the size of this graph?
 - c. Write down the adjacency matrix for this graph.
 - d. Write down the degree sequence for this graph.
 - e. Verify that the sum of the degrees is equal to twice the size of the graph.
2. A simple graph has set of vertices $\{A, E, L, M, N\}$. Two vertices are adjacent in the graph if and only if the letters appear next to each other in the word ANALEMMA.
 - a. Draw this graph.
 - b. Show that this graph is isomorphic to the graph in Question 1 by providing a mapping between vertices which preserves adjacencies.
3. In Questions 1 and 2, suppose that the graphs did not need to be simple. How might they look different? Will these necessarily be isomorphic?
4.
 - a. Show that any two simple graphs with degree sequence 4, 4, 3, 3, 2 are isomorphic.
 - b. Draw a connected non-simple graph with degree sequence 4, 4, 3, 3, 2.
 - c. Draw a disconnected non-simple graph with degree sequence 4, 4, 3, 3, 2.
5. Explain why there are exactly two non-isomorphic simple graphs with the degree sequence 3, 2, 2, 2, 1, and draw an example of each.
6. For the graph shown below, provide an example of:
 - a. A path consisting of at least four vertices;
 - b. A trail consisting of at least four edges that is not a path;
 - c. A walk that is not a trail;
 - d. A Hamiltonian cycle;
 - e. A Eulerian circuit.

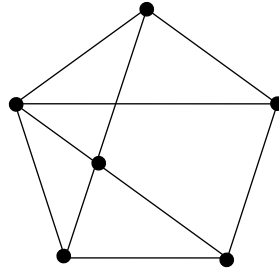
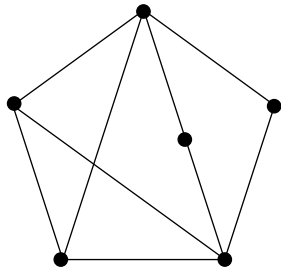


7. For the graph shown below:
 - a. find a Hamiltonian path;
 - b. explain why it is not possible to find a Hamiltonian cycle;
 - c. determine how many different Hamiltonian paths there are.

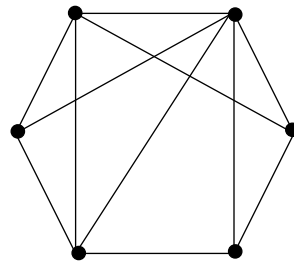
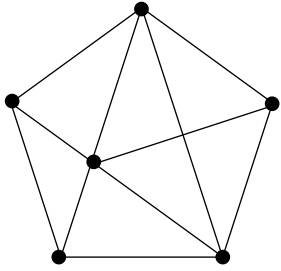


8. Decide whether the following pairs of graphs are isomorphic, proving your assertions.

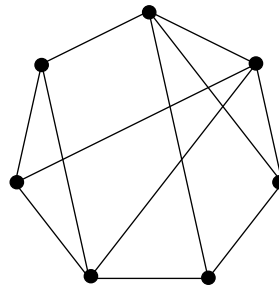
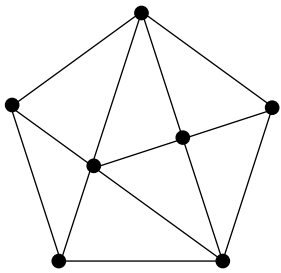
a.



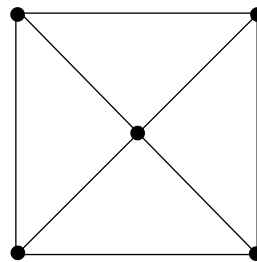
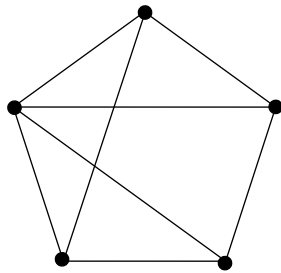
b.



c.



d.



e.

